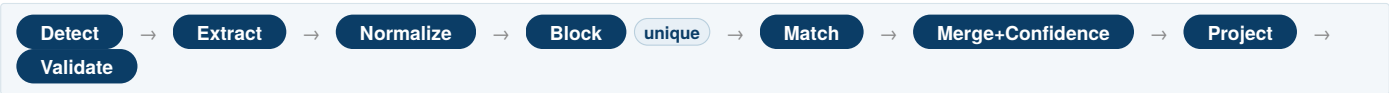


Eightfold Candidate Profile Transformer — Technical Design

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1. Pipeline — Detect → Extract → Normalize → Block → Match → Merge/Confidence → Project → Validate



Detect: sniff source type by content, not filename — never trust extension alone. **Extract:** per-type extractor (CSV/JSON direct map; resume via section-header parsing; notes.txt via lightweight pattern/NER extraction) into a raw intermediate form. **Normalize:** phone→E.164, date→YYYY-MM, country→ISO-3166 alpha-2, skills→canonical via external dictionary. **Block:** group candidates by cheap keys (normalized email/phone/name) before comparison — avoids $O(n^2)$ pairwise comparison, the standard entity-resolution technique enabling **scale to thousands**. **Match:** tiered key resolution (below). **Merge+Confidence:** field-level winner selection + computed confidence. **Project:** apply runtime config on a copy of the canonical record. **Validate:** check projected output against requested config-schema before returning.

2. Canonical Schema & Normalized Formats

All fields from the provided default schema are retained unchanged. Refinements added:

- **candidate_id:** hash(email) → hash(phone) → hash(name)+flag, tiered fallback for identity-less records.
- **years_experience:** *stated* (explicit) vs *derived* (summed from experience date ranges) — tagged by method, never conflated.
- **data_quality_flags[]:** e.g. no_stable_identifier, single_source — makes weak records visibly weak, not just numerically low.
- **provenance[]:** extended to {field, source, method, confidence} — per-entry confidence, not one opaque global score.

Type	Format
Phone	E.164 (+91...)
Date	YYYY-MM (YYYY if month unknown, flagged)
Country	ISO-3166 alpha-2
Skills	Canonical name via external JSON dictionary (e.g. Node.js/NodeJS→Node.js)
Name/Email	Title Case / lowercase, trimmed

3. Merge & Conflict-Resolution Policy

Match keys (tiered, strongest-first):

- T1: exact normalized email — very high confidence
- T2: exact normalized phone — high
- T3: fuzzy email (edit-dist ≤2) + same name — medium
- T4: fuzzy name (Jaro-Winkler ≥0.9) + same employer — low-medium
- T5: no reliable key → kept standalone, **never auto-merged on name alone**

Winner selection (per field type):

- Identity fields: Structured (CSV/ATS) > Resume > Notes
- Current role/title: most-recent source > Resume > CSV
- Experience/education: Resume > Notes > CSV/ATS
- Skills: **union, not override** — agreement raises confidence
- Tie-break: majority agreement, then recency. Near-identical strings (similarity>0.8) treated as same value, not a false conflict.

field_confidence = 0.5×source_reliability + 0.4×cross_source_agreement_ratio + 0.1×recency | overall_confidence = importance-weighted mean (identity fields weighted ~2× skills/headline)

4. Runtime Custom-Output Config (Projection + Validation)

Canonical record is **never mutated** by config — projection runs on a copy, output flows one-way: canonical → **projector** → **validator** → output. **Projector uses one generic path-resolver** (interprets from as a dot/bracket path, e.g. skills[].name) instead of per-field hardcoding — new configs need zero code changes. Supports: field subset selection, rename/remap via from, per-field normalize (registry of normalizer functions), include_confidence/include_provenance toggles, on_missing: null/omit/error. Rule: **required:true always overrides on_missing** — produces a scoped, per-candidate validation error without halting the batch. Validator checks required-fields-present and type-correctness against the requested config-schema before returning.

5. Edge Cases (5) & Deliberate Scope Cuts

Edge Case	Handling
No email/phone at all	Fallback to name-hash ID; flagged no_stable_identifier; capped confidence; never auto-merged on name alone
Near-duplicate identity (slightly different email)	Tier-3 fuzzy match; merge logged with separate match_confidence for auditability
Semantically-equal but textually different values	Similarity check before flagging as conflict; avoids noisy false-conflict logs
Missing structured field (e.g. blank phone)	Left null, never fabricated; that field's confidence is 0, not averaged away
Garbage/corrupt source file	Per-source try/catch; failed source logged & skipped; rest of batch unaffected

Deliberately scoped: live GitHub/LinkedIn API calls (kept pipeline deterministic & offline-runnable); ML-based fuzzy matching (used explainable string-similarity instead, for defensibility); review-queue UI; non-Latin name transliteration. CLI only, per stated lower priority.

Submission: design.pdf (this doc) + public GitHub repo (code, README, sample I/O, tests) + ~2 min demo video (default config run, custom config run, one design decision, one edge case).